RedMeat-TopMed-Metabolites - temporary Quarto to meet ASN deadline

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Important notes

- You should always check you have the most recent files! I use dynamic dates, so this can be done by checking that your files match the versions on GitHub:
  - My profile can be found here
  - My repositories are listed here
  - The most recent version of the source code for this README file can be found here
  - The most recent source code for creating the output files can be found here

Summary

Data Preparation

Metabolites

I will add this later... it's complicated why.

Build Traits

Input Files

- Exam 1 (baseline) covariates and outcomes MESAe1FinalLabel02092016.dta
- Exam 5 covariates and outcomes MESAe5_FinalLabel_20140613.dta
- Exam 6 covariates and outcomes MESAe6_FinalLabel_20220513.dta
- Exam 1 (baseline) energy intake MESAe1_Nutrients_20151208.dta
- Exam 5 energy intake MESAe5_Nutrient_20140416.dta
- Exam 1 (baseline) red meat intake MESAe1_FFQ_20160520.dta
- Exam 5 red meat intake MESAe5_FFQ_20140130.dta

Coding

Predictors

- Red meat was intake of unprocessed red meat in servings / day

Outcomes

- Insulin was measured in mU/L
- Glucose was measured in mg/dL
- HDL was measured in mg/dL
- LDL was measured in mg/dL
- Triglycerides was measures in mg/dL

Covariates

- Race was coded as a factor variable; coded 1= European-American, 2 = Chinese-American, 3 = African-American, 4 = Hispanic-American
- Site was coded as a factor variable; 1 = Wake Forest, 2= Columbia, 3= Johns Hopkins (Baltimore), 4= Minnesota (Twin Cities), 5= Northwestern (Evanston. IL), 6 = UCLA (Los Angeles, VA)
- Sex was coded as a factor variable; coded 1 = female, 2 = male
- Age was in years
- Smoking was a ordinal variable: 0 = Never, 1 = Former, 2 = Current
- Physical activity (PA) is total moderate of vigorous physical activity in Met-Min / week
- Caloric intake (energy) is calories / day
- Kidney functioning is estimated by estimated glomerular filtration rate (eGFR)

Outcomes

- Diabetes was binary; 1 = no diabetes, 2 = diabetes according to ADA 2003 criteria (fasting glucose, use of medication, self-reported diagnosis) and so includes both treated and untreated diabetes
- Cognitive status was ordinal; [1 = no diabetes, 2 = diabetes according to ADA 2003 criteria (fasting glucose, use of medication, self-reported diagnosis) and so includes both treated and untreated diabetes]to fill in]

Build Final Sample

I need to add this too - trying to rush you the actual results ahead of ASN…

Methods

Measures

Metabolome-wide metabolites

Metabolome-wide metabolite profiling was conducted in fasted plasma samples using four complementary liquid chromatography tandem mass spectrometry (LC-MS)-based methods that couple distinct metabolite extraction and LC techniques with high resolution, accurate mass (HRAM) MS profiling to provide a hybrid analysis of hundreds of metabolites of confirmed identity and thousands of LC-MS peaks from yet to be identified compounds. Each of the four LC-MS methods is established on dedicated instruments consisting of Shimadzu Nexera UHPLCs coupled to Thermo Q Exactive series orbitrap mass spectrometers and includes measurement of (i) polar and nonpolar lipids using C8 reversed phase LC and positive ion MS (C8-pos); (ii) free fatty acids, bile acids, and metabolites of intermediate polarity using C18 reversed phase LC and negative ion MS (C18-neg); (iii) amino acids, amino acid metabolites, acylcarnitines, dipeptides, and other cationic polar metabolites using hydrophilic interaction liquid chromatography and positive ion MS (HILIC-pos); and (iv) sugars, organic acids, purines, pyrimidines, and other anionic polar metabolites using HILIC and negative ion MS (HILIC-neg).

Data Processing and compound identification: Targeted data processing of known metabolites was achieved using TraceFinder (v 3.3, Thermo Scientific) software, utilizing an in-house library of over 3000 authentic reference standards with confirmed identities of >700 metabolites (MSI Level 1 ID308). Mixtures of reference standards and reference samples are included in each analysis queue to confirm IDs in every dataset. HRAM, nontargeted data are processed using Progenesis QI software (Nonlinear Dynamics) to detect peaks, perform chromatographic retention time alignment, and integrate peak areas. Metabolites of confirmed identity are then annotated in the dataset, and unknowns are "tagged" using their measured mass-to-charge ratios (m/z) and chromatographic retention times (RT).

Unknowns of interest were initially identified via acquisition of product ion spectra (MS/MS) using a Thermo Scientific ID-X coupled with machine learning-based structure prediction using the SIRIUS/CSI:FingerID workflow and similarity searching using GNPS309, 310. Identities were then confirmed using synthetic standards.

Quality control: During aliquoting, each sample was visually inspected and a pooled QC sample created. LC-MS system performance was qualified prior to analyses using repeated analyses of pooled QCs. During sample analyses: i) mixtures of up to ~150 synthetic reference compounds were analyzed before and after each sample queue to assess consistency of peak shapes, RTs, and signal intensity; ii) internal standards are added to every sample to assure accurate sample volume injection and to monitor drift in sensitivity, and iii) and pairs of pooled QC samples are analyzed at intervals of 20 study samples (10% frequency overall), with the first sample from each pair used for drift correction and the second used to evaluate analytical precision.

Out of >700 identified plasma metabolites and lipids, XXX were measured with a coefficient of variation (CV) < 10%. Similarly, >XXXX peaks from unknowns are consistently measured with a CV <10% in each LC-MS method.

Statistics

All analyses were conducted using R software (version 4.0.5)

Participant characteristics

Demographic, dietary and health information were each calculated as a mean (+/- standard deviation; SD) for continuous variables, or total number (unweighted N) and percentage (%) for categorical or ordinal variables.

Metabolome-wide Associations with Red Meat Intake

Observed associations

To identify metabolites associated with red meat intake, we first estimated the observed red meat-metabolite associations using data from exam 1 and exam 5 within a penalized multi-level model with a random intercept, nested within person, conducted via LASSO with a 50/50 training:test split. LASSO (L_1 regularization) was chosen to maximize sparsity (L_1), given the high-dimensional nature of the data (i.e., the number of predictors was > the number of participants) . At level 1, red meat intake in servings / day was specified as the outcome, with all metabolome-wide metabolites as predictors, alongside the following covariates: age, kideny function (eGFR), diabetes status, income level, physical activity, highest education level achieved, average daily energy intake, gender, time point (exam 1 / exam 5), race\ethnicity, and smoking status, which was nested within person (level 2). A penalty parameter was added (λ), selected via 5-fold cross-validation to optimize predictive performance, and the optimal λ applied to metabolite coefficients only (i.e., not covariates), shrinking weak or redundant metabolite predictors toward zero and mitigating over-fitting and multi-collinearity among highly correlated metabolites.

Age, physical activity, estimated glomerular filtration rate, daily energy intake, diabetes status, smoking status, data collection site and exam included as time-varying covariates, and sex, race\ethnicity, and income level included as time invariant covariates. All non-categorical variables (metabolite predictors, red meat outcome, and covariates) were scaled and standardized (z-score standardization to a mean of 0 and a standard deviation of 1) prior to inclusion in the models.

Within-Person Effects

Next, we identified metabolites that covaried with changes in red meat intake within individuals via centering to isolate within-person effects which reflect the covariation of the predictor and outcome within individuals over time. These capture statistical responses of the outcome to change in the predictor and may reflect causal effects, but can also be influenced by time-varying confounding (i.e., influences that covary with both the predictor and the outcome) and measurement error. First: the person-mean for red meat intake was calculated, that is, the mean red meat intake across both time points for each participant. For the ith person, this is derived using the equation:

$$\bar{X}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} X_{ij}$$

where X_{ij} is the observation of red meat intake for the ith person at the jth time point. This removes any within-person variance and represents each participant's average red meat intake during the course of study, and by definition only contains between-person variance (plus possible measurement error).

Next, the observations of red meat intake are person centered, by subtracting the person mean from each observation. For the ith person, at the jth time point, this is derived using the equation:

$$X_{ij}^{\text{CWC}} = X_{ij} - \bar{X}_i$$

where X_{ij} is the observation of red meat intake for the ith person at the jth time point. This centering approach removes any between-person variance in red meat intake, with each observation representing the participant's difference in intake compared to their own usual intake, and thus by definition only contains within-person variance.

This person centered variable was then used as the outcome in penalized (again, LASSO penalty) multi-level models, otherwise specified as above.

Creation of metabolite summary scores

For each parameter (observed associations [β_{Obs}], and within-person effects [β_{WP}], a weighted sum of all metabolites selected by the penalized model for that parameter was created at each of exam 1, 5 and 6, by multiplying individual standardized metabolite values by the relevant coefficient from the penalized model. Thus, two scores were created for each participant at each time point: (1) a score representing the sum of significant metabolites from the observed association analyses ($Score_{Obs}$); and (2) a score representing the sum of significant metabolites from the within-person effects ($Score_{WP}$).

Associations Between Metabolite Scores and Health

We examined the associations between the two metabolite scores and health using data from exam 6, to decouple the data used for these analyses from the data used to generate the metabolite scores. Since we were restricted to using data from one exam only, we examined observed associations between each of four metabolite scores, and each of cardiometabolic disease risk factors using standard linear regression models, with the metabolite scores as the predictor and risk factor as the outcome, which was transformed using an inverse normal transformation prior to inclusion in the models. The exception was for incident diabetes, for which metabolite scores from baseline were used for all without T2D at this exam. All models included the same set of covariates used in the penalized models, specified as fixed effects.

Results

Sample descriptives

- Sample descriptives at baseline are available in Table 1

Table 1: Sample Descriptives, Stratified by Exam

	Exam	
Characteristic	**1** N = 6,171[1]	**5** N = 4,273[1]
Age (y)	62.28 (10.24)	69.76 (9.40)
Gender		
Female	3,225 / 6,171 (52%)	2,245 / 4,273 (53%)
Male	2,946 / 6,171 (48%)	2,028 / 4,273 (47%)
Race or ethnicity		
Non-Hispanic White	0 / 0 (NA%)	0 / 0 (NA%)
Chinese American	0 / 0 (NA%)	0 / 0 (NA%)
Black/African-American	0 / 0 (NA%)	0 / 0 (NA%)
Hispanic	0 / 0 (NA%)	0 / 0 (NA%)
Smoking status		
Never	0 / 0 (NA%)	0 / 0 (NA%)
Former	0 / 0 (NA%)	0 / 0 (NA%)
Current	0 / 0 (NA%)	0 / 0 (NA%)
Diabetes status		
Normoglycemia	5,397 / 6,159 (88%)	3,401 / 4,226 (80%)
Diabetes	762 / 6,159 (12%)	825 / 4,226 (20%)
Physical activity, (MET-min/week)	936.28 (2,823.60)	778.84 (2,706.43)
Estimated glomerular filtration rate	80.89 (18.42)	79.98 (21.05)
Energy intake, (kcals/day)	1,624.24 (864.65)	1,630.59 (1,090.48)
Red meat intake, (svd/day)	0.43 (0.43)	0.42 (0.57)
Triglycerides (mg/dL)	132.27 (87.36)	109.35 (60.75)
LDL cholesterol (mg'dL)	117.18 (31.40)	105.05 (32.26)
HDL cholesterol (mg/dL)	50.96 (14.81)	55.86 (16.76)
Fasting glucose (mg/dL)	97.26 (30.39)	101.99 (28.59)
Fasting insulin (mu/mL)	10.37 (16.91)	59.75 (48.19)
Cogntive status at follow-up		
No impairment	0 / 0 (NA%)	0 / 0 (NA%)

Mild Ccognitive impairment	0 / 0 (NA%)	0 / 0 (NA%)
Probable dementia	0 / 0 (NA%)	0 / 0 (NA%)

[1]Mean (SD); n / N (%)

Metabolite Selection

First, we ran a penalized model to identify an optimal set of metabolites that jointly predicted red meat intake at the population level. The full results are available here. From N=2811 metabolites, N=153 metabolites were selected by the penalized model. A file of all metabolites selected by the penalized model for red meat intake at the population level, with their associated coefficients is available here. Table 2 shows those metabolites, plus their respective coefficients, selected by the penalized model as associating with red meat intake at the population level that were also known - or identified - compounds.

[Matthew, most are lipids, which is common (metabolite panels are enriched for lipids). I think most interesting here are DHA, and EPA - you could give these as examples of fatty acid associations (or inverse associations) with red meat intake in the population level that would lead to negative associations of meat with health, but are not reflected in the within-person effects]

Post ASN - Lekki to add a table of the linear regression results

First, we ran a penalized model to identify an optimal set of metabolites that jointly predicted the within-person effects of red meat intake. The full results are available here. From N=2958 metabolites, N=78 metabolites were selected by the penalized model. A file of all metabolites selected by the penalized model for red meat intake at the population level, with their associated coefficients is available here. Table 3 shows those metabolites, plus their respective coefficients, selected by the penalized model as associating with red meat intake at the population level that were also known - or identified - compounds.

[Matthew, most are lipids, which is common (metaboliote panels are enriched for lipids). I think most interesting here are DHA, and EPA - you could give these as examples of fatty acid associations (or inverse associations) with red meat intake in the population level that would lead to negative associations of meat with health, but are not reflected in the within-person effects]

Post ASN - Lekki to add a table of the linear regression results

Figure 1 shows the overlap in metabolites selected by the penalized model for observed associationat the population level vs. those selected by the model for within-person effects. One metabolite overlapped (see Table 4).

[Matthew - boringly, this was an unknown metabolite, and it had the same direction of effect in both models. Not as exciting as we had hoped… but we just report it honestly. Up to you if you bring it up in your talk, or just say "this was the only compound selected by both models" and leave others to ask if the direction of effect was the same]

Table 2: Metabolites and coefficients, selected by the penalized model as associating with red meat intake at the population level for known compounds.

Original_Metabolite_Name	Estimate
DHA	-6.229331e-03
LPC 22:6/0:0	5.444630e-04
PC 34:5	-8.708483e-06
PC 38:6	-6.110022e-03
PC 38:1	-2.389285e-06
PC 39:6	5.896305e-04
PC 40:9	-4.798044e-03
PC 40:6	-2.166861e-03
PC 42:5	3.833519e-05
PC O-38:6	4.767516e-05
PC P-40:6 or PC O-40:7_B	-2.932681e-05
PC P-44:6 or PC O-44:7	-1.109323e-02
PC P-46:6 or PC O-46:7	-8.742721e-03
PS 34:0	2.649808e-04
SM 44:3;O2	9.707168e-05
SM 44:2;O2	1.263018e-04
CE 22:6	-1.207683e-03
TG 52:8	-3.896299e-07
TG 52:7	-1.561114e-05
TG 54:9	9.594103e-05
TG 54:8	1.216304e-05
TG 55:6	1.951195e-04
TG 56:10	1.335861e-05
TG 56:9	3.817295e-04
TG 56:8	4.955870e-04
TG 56:7	4.585226e-04
TG 58:11	1.250745e-04
TG 58:10	2.804596e-04
TG 58:9	3.069765e-04
TG 58:7_A	1.700680e-06
TG 60:12	2.022498e-04
Eicosapentaenoic acid	1.398778e-04
Docosahexaenoic acid	-6.530131e-03
Docosapentaenoic acid (22n-6)	2.979272e-04
11-methyl lauric acid	2.791778e-04
CAR 13:0	1.175697e-04
CAR 16:4	3.533226e-05
PC P-36:5 or PC O-36:6	4.071590e-04
PC P-35:1 or PC O-35:2	3.667091e-04
PC P-38:6 or PC O-38:7	5.998464e-04
LPE 22:6	-1.241959e-02
PE P-35:2 or PE O-35:3	4.313000e-03
PE P-38:6 or PE O-38:7	6.336840e-04
PE P-38:4 or PE O-38:5	4.272790e-04
PE P-40:6 or PE O-40:7	2.860782e-04
PE 38:6	3.798159e-04
PE 40:6	3.426626e-04

Table 3: Metabolites and coefficients, selected by the penalized model as showing within-person associations with red meat intake for known compounds.

Original_Metabolite_Name	Estimate
LPC 17:0/0:0	-0.0006055623
TG 54:3	-0.0006508867

Table 4: Metabolites and coefficients, selected by both penalized models

Metabolite	Population_level_Coefficient	Within_Person_Coefficient
C8_QI1850	0.000619385	0.05896401

NULL

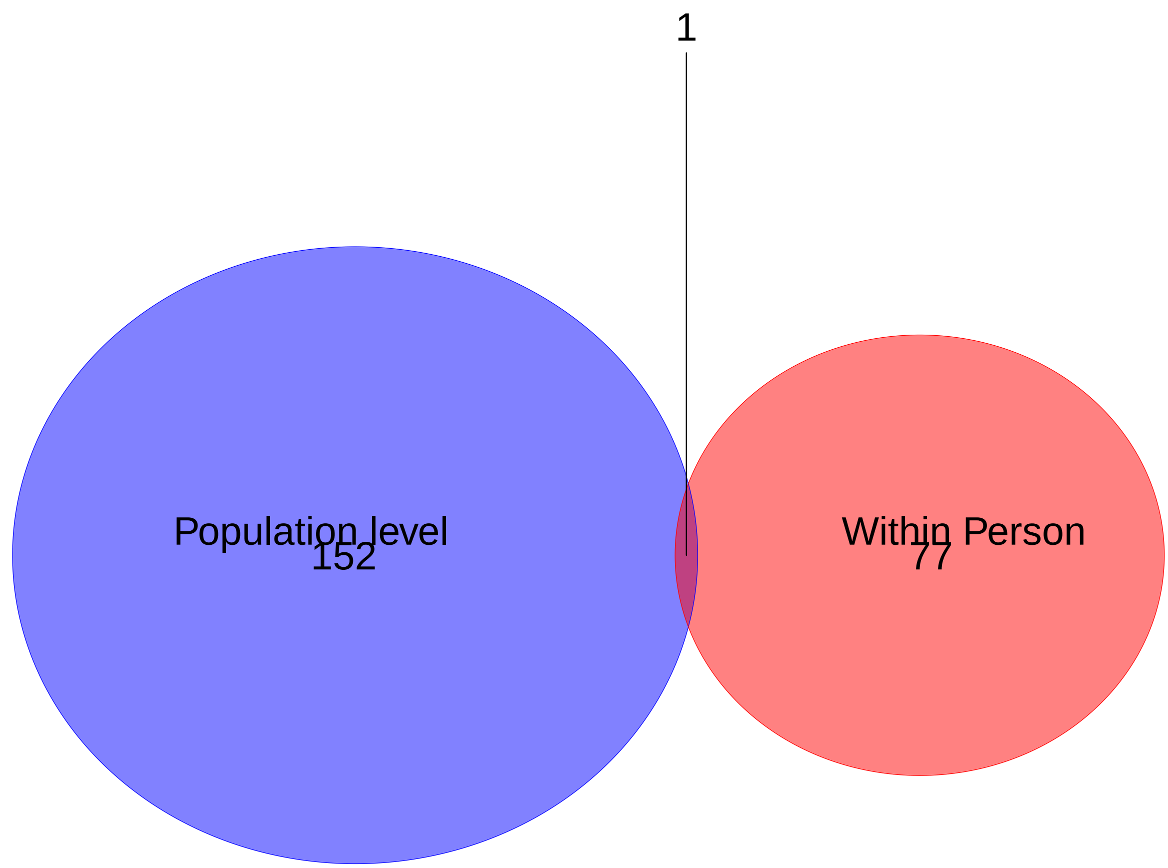

Figure 1: Overlap in XXXXs

Parameterizing Red Meat - Metabolite Associations

LASSO coefficients can be tricky to interpret as they represent a combination of variable selection and shrinkage nshinkage is achieved via the L_1 penalty(. Because of this, you generally **do not interpret Lasso coefficient magnitudes** as the literal effect of X on Y. Instead, they represent a *penalized* estimate used specifically to optimize out-of-sample prediction, and select an optimal set of metbaoloites that jointly predict the red meat variable of interest, without overfitting. To help understand the magnitude of associations for the metabolites selected by the penalized models, for all metabolites selected by the penalized model, we also fit standard multi-level models without a penalty term, specified in the same manner as for the penalized models, but with each metabolite specified as an independent predictor.

For the population-level associations, the full results can be found here. Of the N=153 metabolites selected by the penalized model, the associations of N=100 metabolites were statistically significant at P<.05. The associations of a further N=90 remained statistically significant after an FDR-correction for multiple testing.

For the within-person associations, the full results can be found here. Of the N=153 metabolites selected by the penalized model, the associations of N=14 metabolites were statistically significant at P<.05. The associations of a further N=1 remained statistically significant after an FDR-correction for multiple testing.

[Matthew… this is not ideal.. it speaks ot a lack of power… but it is what it is at this point… this will probably be supplementary material in the paper]

We also discussed interest in seeing whether the metabolites selected by one penalized model, but not by a second penalized model, truly had differential population-level associations vs. within-person effects. I have put this information here.

[Matthew: When we write the final paper we can talk about how to interpret this. It is a plausible and complex pattern of associations, and we need to decide a clean “take home message” re: red meat and health, and how robust our analyses are. I have an opinion, but I am interested in your thoughts (after ASN…)].

Associations of Metabolite scores with health

Table 5 shows the associations of the two metabolites with our health measures.

Table 5

	Population Level Association			Within-Person Effect		
Characteristic	**Beta**	**SE**	**p-value**	**Beta**	**SE**	**p-value**
Fasting Glucose	0.076	0.010	1.83×10^{-14}	-0.020	0.007	4.69×10^{-3}
Fasting Insulin	0.210	0.011	4.41×10^{-85}	-0.057	0.009	3.94×10^{-11}

Abbreviations: CI = Confidence Interval, SE = Standard Error